CAPT: Cross-modal Anchored Prompt Tuning
for Universal Cross-Domain Retrieval

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Motivation

Universal Cross-Domain Retrieval (UCDR) retrieves relevant images when the query comes from an **unseen domain**, an **unseen class**, or both [1]. Recent CLIP-based methods [3–6] learn or simulate prompt components from the *source* domains, which can entangle class semantics with the visual style of the training domains.

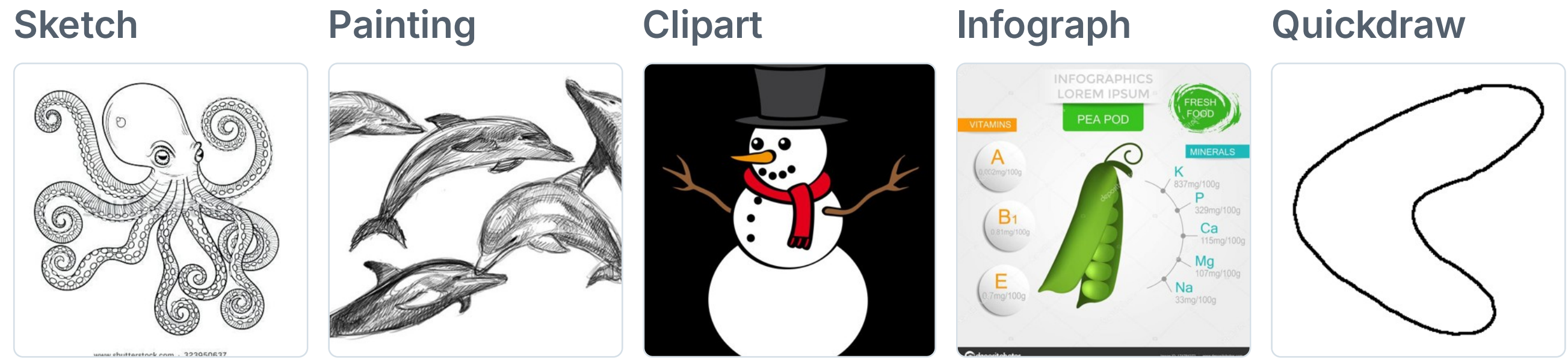
We adapt a fully frozen CLIP [2] **without any domain prompt**: each image reads out its own visual prompt from CLIP's frozen class-name text anchors.

Contributions

- **DSIP** reads layer-wise, instance-specific visual prompts out of frozen text-class anchors. No domain prompt, no backbone fine-tuning.
- **PACE** routes each sample between contrastive alignment and prototype anchoring by its residual domain gap. No hand-tuned loss weight.
- State of the art on DomainNet, Sketchy and TU-Berlin with **≤1.35M** trainable parameters.

Task

Query (held-out domain, unseen class)


Sketch
Painting
Clipart
Infograph
Quickdraw

CAPT top-1 in the real-photo gallery

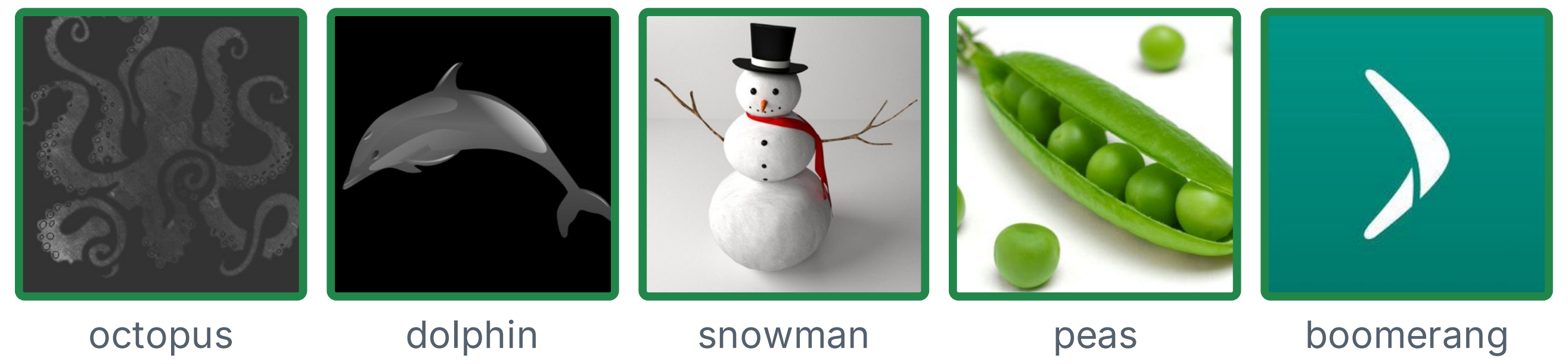

octopus
dolphin
snowman
peas
boomerang

Figure 1. One query per held-out DomainNet domain (the first sampled query whose top-1 is correct).

Unseen	unseen-domain, unseen-class queries
Mixed	same queries; gallery adds seen-class distractors
U^{d}CDR	unseen-domain queries from seen classes

Method

frozen CLIP ViT-B/32, 1.35M trainable

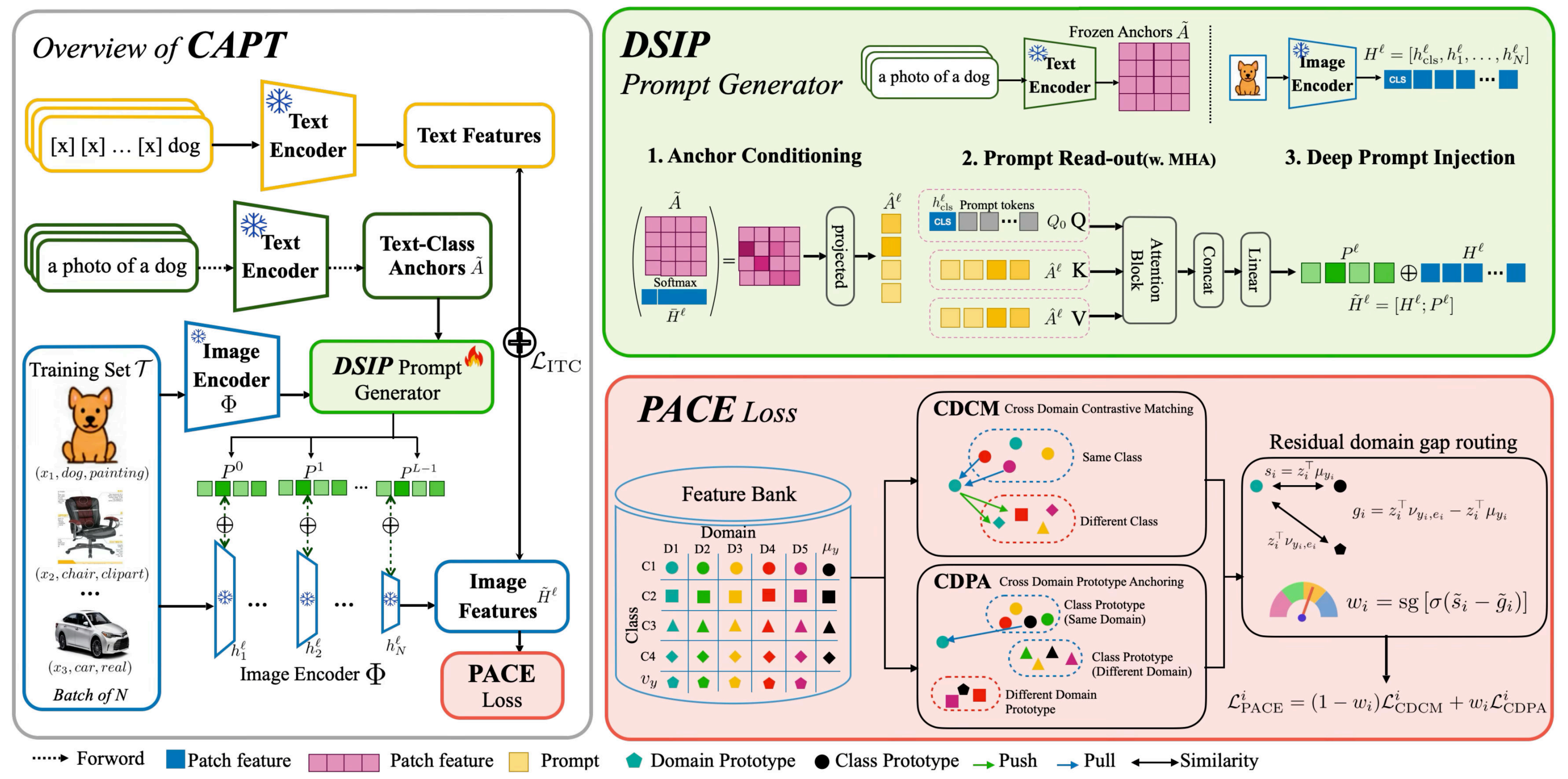

Overview of CAPT
[x] [x] … [x] dog
Text Encoder
Text Features
a photo of a dog
Text Encoder
Text-Class Anchors $\tilde{A}$
Training Set $\mathcal{T}$
$(x_1, dog, painting)$
$(x_2, chair, clipart)$
$(x_3, car, real)$
Batch of N
Image Encoder Φ
DSIP Prompt Generator
$\mathcal{L}_{\mathrm{ITC}}$
P^0
P^1
P^{L-1}
h_1^ℓ
h_2^ℓ
h_N^ℓ
Image Encoder Φ
Image Features $\tilde{H}^\ell$
PACE Loss
DSIP
Prompt Generator
a photo of a dog
Text Encoder
Frozen Anchors $\tilde{A}$
Image Encoder
$H^\ell = [h^\ell_{\mathrm{cls}}, h^\ell_1, \dots, h^\ell_N]$
CLS
1. Anchor Conditioning
2. Prompt Read-out(w. MHA)
3. Deep Prompt Injection
$\tilde{A}$
Softmax
$\bar{H}^\ell$
projected
$\hat{A}^\ell$
h^ℓ_{cls} Prompt tokens
Q_0 Q
$\hat{A}^\ell$ K
$\hat{A}^\ell$ V
Attention Block
Concat
Linear
P^ℓ
H^ℓ
$\tilde{H}^\ell = [H^\ell; P^\ell]$
PACE Loss
Feature Bank
Domain
D1 D2 D3 D4 D5 μ_y
Class
C1 C2 C3 C4 ν_y
CDCM Cross Domain Contrastive Matching
Same Class
Different Class
CDPA Cross Domain Prototype Anchoring
Class Prototype (Same Domain)
Class Prototype (Different Domain)
Different Domain Prototype
Residual domain gap routing
$s_i = z_i^\top \mu_{y_i}$
$g_i = z_i^\top \nu_{y_i,e_i} - z_i^\top \mu_{y_i}$
$z_i^\top \nu_{y_i,e_i}$
$w_i = \mathrm{sg}\,[\sigma(\tilde{s}_i - \tilde{g}_i)]$
$\mathcal{L}^i_{\mathrm{PACE}} = (1 - w_i)\mathcal{L}^i_{\mathrm{CDCM}} + w_i\mathcal{L}^i_{\mathrm{CDPA}}$
Forword
Patch feature
Patch feature
Prompt
Domain Prototype
Class Prototype
Push
Pull
Similarity

Figure 3. DSIP injects prompts read from image-conditioned text anchors into every layer of the frozen image encoder. PACE routes each sample between CDCM and CDPA.

DSIP

Text anchors $\tilde{A}$ (“a photo of a [class]”, seen classes only) attend to the patches of the current image:

$$\hat{A}^\ell = \tilde{A} + \sigma(\gamma)\,\mathrm{softmax}\Big(\frac{\tilde{A}(\bar{H}^\ell)^\top}{\sqrt{r}}\Big)\bar{H}^\ell$$

Static queries Q_0 read an image-specific prompt, appended at all 12 layers:

$$P^\ell = \mathrm{MHA}_\ell(Q_0, \hat{A}^\ell, \hat{A}^\ell)\,W_u^\ell$$

The prompt stays inside the anchor hull (residual <2×10^{-7} at every layer).

PACE

CDCM: same class, other domain is a positive (bank of 10k, 256 hard negatives). **CDPA**: $1 - z_i^\top \mu_{y_i}$. The residual domain gap

$$g_i = z_i^\top \nu_{y_i,e_i} - z_i^\top \mu_{y_i}$$

and confidence $s_i = z_i^\top \mu_{y_i}$ set a per-sample weight:

$$w_i = \mathrm{sg}\big[\sigma(\tilde{s}_i - \tilde{g}_i)\big]$$

$$\mathcal{L}^i_{\mathrm{PACE}} = (1 - w_i)\mathcal{L}^i_{\mathrm{CDCM}} + w_i\mathcal{L}^i_{\mathrm{CDPA}}$$

Domain labels are used only in training.

Training: $\mathcal{L} = \mathcal{L}_{\mathrm{ITC}} + 0.5\,\overline{\mathcal{L}_{\mathrm{PACE}}}$; $P_q = 8$, $r = 64$, Adam 10^{-4}, batch 60, epoch 4. Inference is one label-free forward pass with cosine similarity.

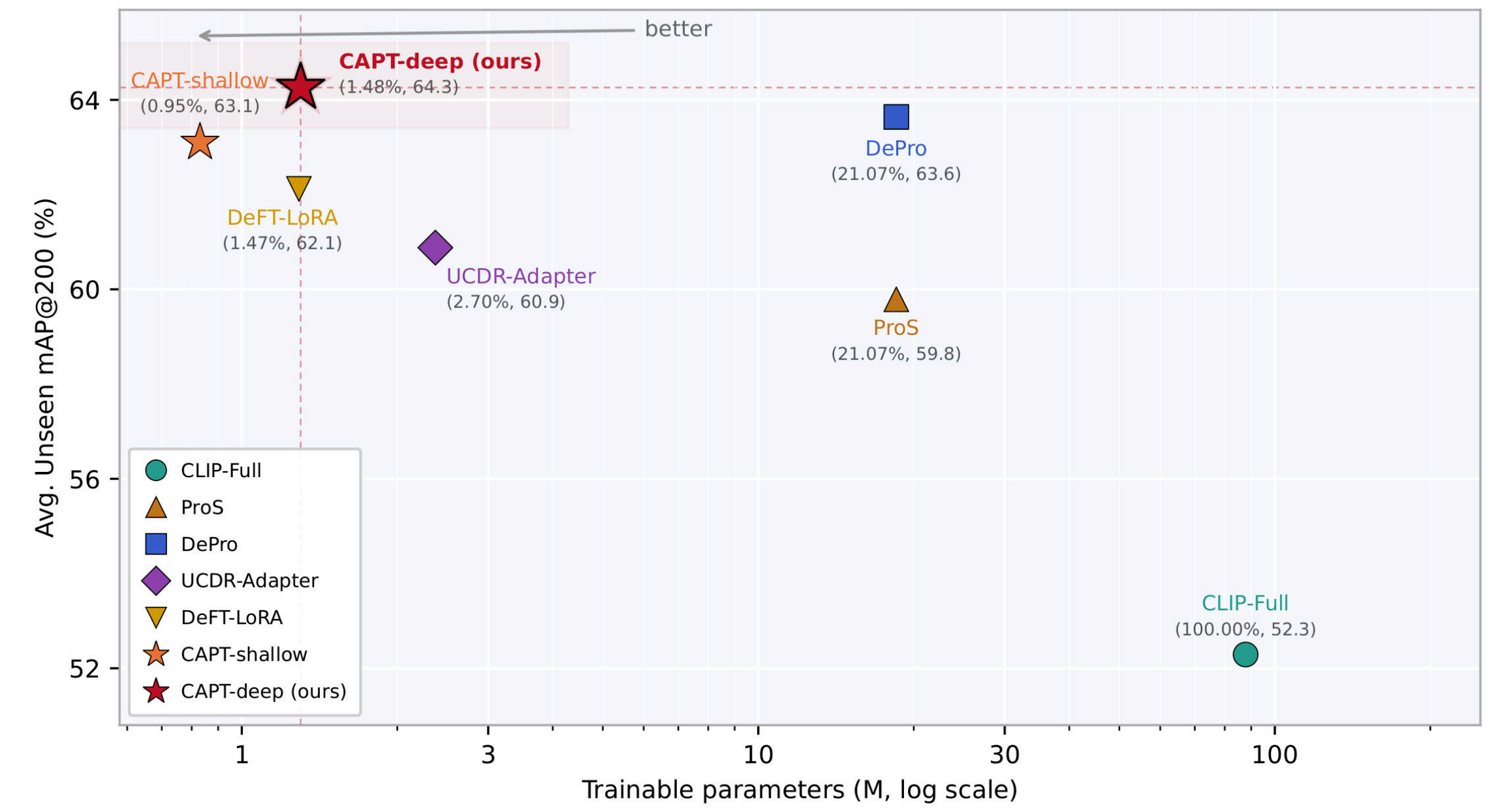


Figure 2. Average Unseen mAP@200 vs. trainable parameters. CAPT trains about 1.35M parameters, on par with DeFT-LoRA (1.29M).

DomainNet

5 held-out domains

Method		Unseen		Mixed		U^{d}CDR	
		mAP	Prec	mAP	Prec	mAP	Prec
ProS†	CVPR'24	.5979	.5550	.5470	.5110	.6345	.3991
UCDR-Adapter†	WACV'25	.6088	.5665	.5666	.5312	.6437	.4067
DeFT-LoRA	AAAI'26	.6212	.5776	.5690	.5331	.6547	.4105
DePro†	SIGIR'25	.6364	.5996	.5686	.5398	**.6881**	**.4282**
CAPT-shallow	ours	.6310	.5913	.5664	.5345	.6616	.4092
CAPT-deep	**ours**	**.6428**±.0010	**.6057**±.0009	**.5753**±.0008	**.5473**±.0008	**.6829**±.0008	**.4240**±.0001

Table 1. Average over the held-out domains, mAP@200 / Prec@200. **Bold** best, underline runner-up. † reproduced. CAPT-deep: mean ± std of three seeds.

Unseen mAP@200 per held-out domain

Method	Clipart	Sketch	Infograph	Painting	Quickdraw
ProS†	.7602	.6451	.5634	.7488	.2718
UCDR-Adapter†	.7719	.6570	.5803	.7547	.2800
DeFT-LoRA	.7739	.6663	.6093	.7551	.3012
DePro†	.7906	.6843	.6174	.7701	.3195
CAPT-shallow	.7850	.6789	**.6201**	.7724	.2984
CAPT-deep	**.7990**	**.6944**	.6200	**.7728**	**.3278**

Against DePro, CAPT-deep wins **21 of 30** cells, including all **20** Unseen and Mixed cells. Margins are modest, and DePro keeps the average U^{d}CDR mAP (.6881 vs. .6829).

Fair frozen backbone

Method	Unseen	Mixed	U^{d}CDR
DePro-Frozen	.6132/.5783	.5426/.5163	.6765/.4257
DePro-LN	.6364/.5996	.5686/.5398	**.6881/.4282**
CAPT	**.6428/.6057**	**.5753/.5473**	**.6829/.4240**
CAPT+LN+bias	**.6458/.6085**	**.5782/.5502**	.6877/.4260

Table 2. DePro's Table 1 entry tunes CLIP's LayerNorm. Fully frozen, CAPT wins 26 of 30 cells; with the same freedom (1.55M parameters) the U^{d}CDR gap shrinks to .0004.

Cross-dataset

Method	Sketchy		TU-Berlin	
	mAP@200	Prec@200	mAP@all	Prec@100
ProS†	.7151	.6739	.6138	.7037
DePro†	.7515	.7177	.6629	.7203
UCDR-Adapter†	.6933	.6490	.6314	.7099
CAPT	**.7638**	**.7291**	**.6641**	**.7305**

Table 3. Sketch-to-image, unseen classes. † our reimplementation, same environment, full Sketchy gallery.

Qualitative

infograph, unseen classes

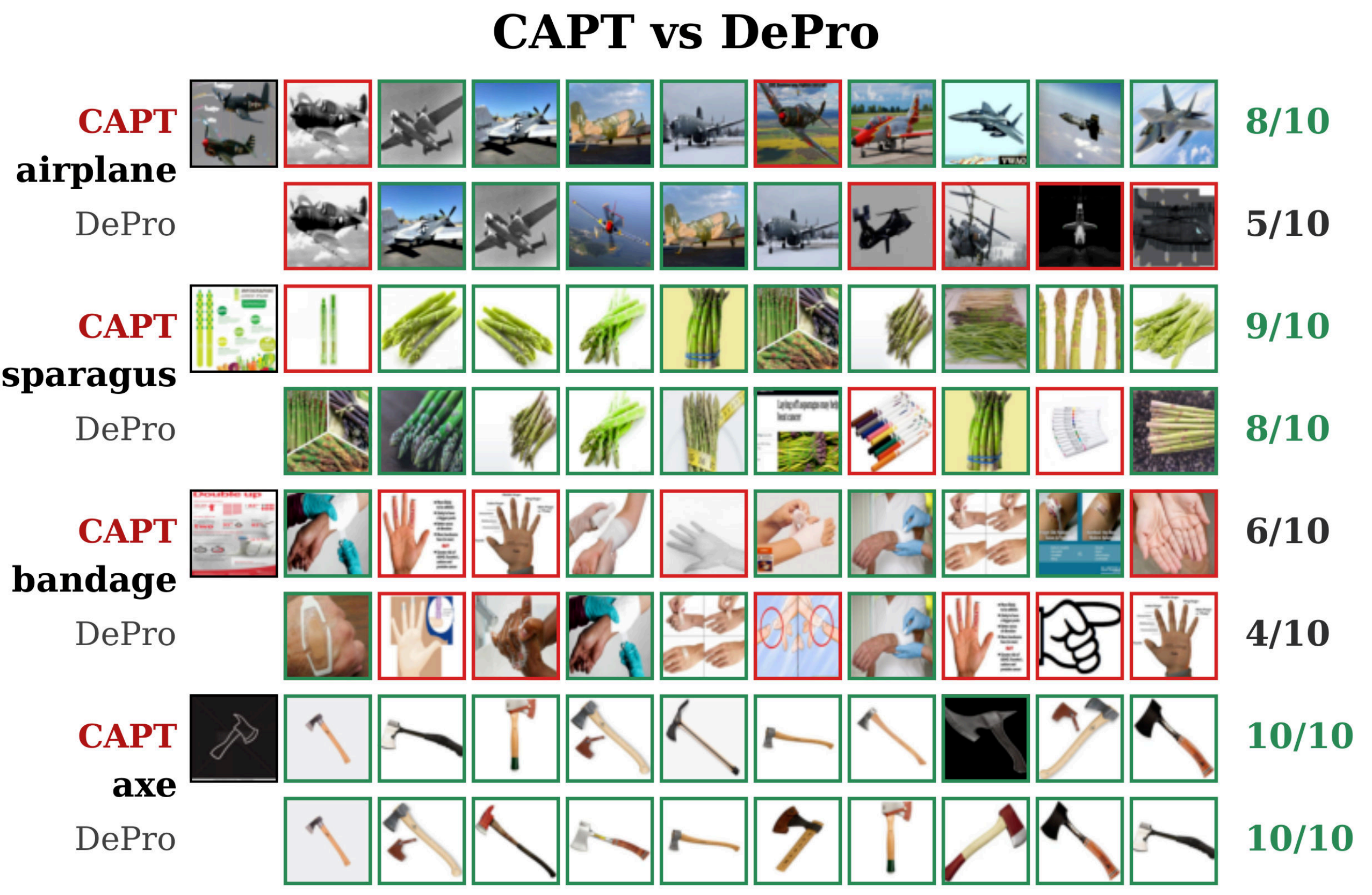

CAPT vs DePro
CAPT
airplane
DePro
8/10
5/10
CAPT
asparagus
DePro
9/10
8/10
CAPT
bandage
DePro
6/10
4/10
CAPT
axe
DePro
10/10
10/10

Figure 4. Top-10 on shared queries; CAPT wins airplane, asparagus and bandage, ties axe.

What matters

Sketch holdout	Unseen	Mixed	U^{d}CDR
Frozen + ITC	.3901	.3391	.4468
+ read-out	.6602	.5867	.7551
+ image-cond. (DSIP)	.6688	.5880	.7669
+ CDCM	**.6770**	**.5981**	.7712
+ PACE (full)	**.6766**	**.5962**	**.7729**

Table 4. Cumulative build-up, mAP@200 (earlier code revision, not matched to Table 1).

- The anchored **read-out** gives most of the gain. Text anchors add a modest prior over a random basis (+.0021).
- No **fixed weight** matches per-sample routing (14–15 of 15 runs), and shuffling w_i within the batch also loses (13 of 15).
- PACE's Unseen gain is not significant (p=.19); its U^{d}CDR mAP gain is (+.0077, p=.005).

Limitations

DePro with LayerNorm tuning stays slightly stronger on a few U^{d}CDR cells, and PACE's prototypes degrade under severe class imbalance. The refinement layer is not a contribution, but without it the tally against DePro drops from 22/30 to 14/30.

1. Paul et al. Universal cross-domain retrieval. ICCV 2021.
2. Radford et al. Learning transferable visual models from natural language supervision. ICML 2021.
3. Fang et al. ProS. CVPR 2024.
4. Jiang et al. UCDR-Adapter. WACV 2025.
5. Chen et al. DePro. SIGIR 2025.
6. Xu et al. DeFT-LoRA. AAAI 2026.